



**UNIVERSITY EXAMINATIONS: 2023/2024**

**EXAMINATION FOR BACHELOR OF EDUCATION ARTS**

**SMA 2002: CALCULUS I**

**DATE: DECEMBER, 2023**

**TIME: 2 HOURS**

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**INSTRUCTIONS: Answer Question One and Any Other Two Questions**

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**QUESTION ONE (20 MARKS)**

- a) Use the first principles of differentiation to find the derivative of, **(4 Marks)**

$$f(x) = \frac{3x}{1 - 5x}$$

- b) If  $f(x) = x^2 + 3$  and  $g(x) = \sqrt{x}$ . Find  $(f \circ g)x$  and  $(g \circ f)x$ . **(4 Marks)**

- c) Evaluate **(3 Marks)**

$$\lim_{x \rightarrow \infty} \frac{5x^2 - 3x + 2}{10x^2 - x + 100}$$

- d) Find the derivatives of the following curves

- e) i)  $y = 5x^4 + 4x - \frac{1}{2x^2} + \frac{1}{\sqrt{3}} - 3$  **(3 Marks)**

- ii)  $y = (2x - 1)^6$  **(3 Marks)**

- iii)  $y = (1 - 5x^2)(6x^2 - 4x + 1)$  **(3 Marks)**

**QUESTION TWO (15 MARKS)**

- a) Find the derivative with respect to  $x$  **(5 Marks)**

$$y = x^x \sin^{-1} 2x$$

- b) Find an equation of the tangent to  $x^2 + xy + 2y^3 = 4$  at  $(-2, 1)$ . **(5 Marks)**

- c) Find  $\frac{dy}{dx}$  given  $x = e^t \cos 2t$  and  $y = e^{-t} \sin 2t$ . (5 Marks)

### QUESTION THREE (15 MARKS)

- a) Find the maximum and minimum values of the curve  $y = x^3 - 3x + 5$ . Hence sketch the graph. (7 Marks)
- b) An open box is to be constructed with a square base and is required to have a volume of 48 cubic inches. The bottom of the box costs 3 cents per square inch, whereas the side cost 2 cents per square inch. Find the dimension that will minimize the cost of the box.

(8 Marks)

### QUESTION FOUR (15 MARKS)

- a) Find  $a$  and  $b$  so that the following functions are continuous  $\forall x \in \mathbb{R}$

(4 Marks)

$$f(x) = \begin{cases} 2, & \text{if } x < 1 \\ ax + b, & \text{if } 1 \leq x < 2 \\ 6, & \text{if } x \geq 2 \end{cases}$$

- b) Find the derivative with respect to  $x$  of the following functions.

i.  $y = \frac{x}{\tan x}$  (3 Marks)

ii.  $f(x) = \sinh^{-1}(\ln x)$  (4 Marks)

- c) Show that  $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ , hence find  $\lim_{\theta \rightarrow 0} \frac{\sin 2\theta}{5\theta}$  (4 Marks)

### QUESTION FIVE (15 MARKS)

- a) Using  $y = \sqrt[3]{x}$ , find the approximate value of  $\sqrt[3]{126}$ . (4 Marks)
- b) The parametric equations of a curve are  $x = e^t, y = \sin t$ . Find  $\frac{d^2y}{dx^2}$  as a function of  $t$ . Hence show that  $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$ . (7 Marks)
- c) The distance  $x$  metres moved by a car in time  $t$  seconds is given by

$$x = 3t^3 - 2t^2 + 4t - 1. \text{ Determine the velocity and acceleration when } t = 1.5s.$$

**(4 Marks)**